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COURSE ASSESSMENT 2024 FOR GROUP ASSIGNMENT/PRESENTATION
CTE711S COMPILER TECHNIQUES

COURSE DETAILS

Course Code:	CTE711S	Semester, Year:	Sem 1, 2026
Course Title:	Compiler Techniques		
Lecturers:	Coordinator:	Prof A. A. Azeta	
Title of Assessment:	Development of a Mini Compiler software using Java		

SUBMISSION

Soft Copy:	• Students should divide themselves into groups of minimum 4 and maximum 5 per group. • The Assignment should be done as a group, but each student should submit a copy of the documentation in MS Word file on elearning. • Each file should be giving the name and student number (i.e. Steven-984322220911). • All Submissions of documentations must be done on e-Learning on or before 23h59, 1 May 2026. • Date of Group Assignment presentation/defence via Face-to-Face (F2F) Will be 4 – 8 May 2026.	
Hard Copy:	Not required (If required will be communicated)	
Late Submission:	Late submission will not be entertained without prior approval	
Multiple Hand-ins:	Multiple hand-ins before the due date is allowed but the latest version will be graded.	
Assessment	**20%**	

Further information

1. EXTRA MARKS WILL BE AWARDED FOR CRITICAL THINKING, CREATIVE IDEAS AND MOST PRACTICAL APPLICATION
2. PLEASE NOTE THAT INABILITY TO EXPLAIN YOUR CODE WILL LEAD TO ALL THE MARKS BEING PEGGED AT ZERO (SEE TEACHING PHYLOSOPHY IN COURSE OUTLINE).
3. PLAGARISM AND AI GENERATED CONTENT WILL NOT BE TOLORATED AND HENCE IF ASSIGNMENTS ARE COPIED OR SHARED, THEN ALL STUDENTS INVOLVED ARE DISQUALIFIED.

GROUP ASSIGNMENT DETAILS

The Program below is written in V language to check for errors, evaluate and write out M.

```
BEGIN
INTEGER A, B, C, E, M, N, G, H, I, a, c
INPUT A, B, C
LET B = A */ M
LET G = a + c
temp = <s%**h - j / w +d +*$&;
M = A/B+C
N = G/H-I+a*B/c
WRITE M
WRITEE F;
END
```

The Grammar and Production rules:

E→E	**Rule1 (R1)**
E→|E+E|E/E|E*E|E+E|	**R2**
E→|E1|E2|E3|…|E26|	**R3**
[E1|E2|E3|…|E26|]→{|A|B|C|…|Z|a|b|c|…|z|}	**R4**

Note the following additional conditions in the above program:

- Words in capital letters are **Keywords**
- Words in small letters are **Identifiers**. Words in single letters from A – Z and a to z are also **Identifiers**
- +, -, /, * are **Operators**
- Equals =, and semi colon ; are **Symbols**
- Any string/line must contain: **Keywords, Identifiers, Operators, or Symbols**
- Symbols such as: %, $, &, <, >, ; not allowed and would give **Semantic error**
- Two operators must not be combined such as: +* not allowed and would give **Syntax error**, same with other operators that are combined, for example -/, */, *+
- Semi colon ; at the end of a line not allowed and would give **Syntax error**
- Numbers 0,1 to 9 are not allowed and would give **Syntax error**
- The acceptable keywords are: BEGIN, INTEGER, LET, INPUT, WRITE, END.
- Misspelling in the keywords such as: WRITEE not allowed and would give **Lexical error**
- Any other character on the keyboard different from all the above will give Syntax error.
- Note that text with yellow background in the above program have errors

(1) **Write a Java program that will go through the following stages of compiler to translate the above program (iteratively) - line by line.**

(2) **Write a Java program that will go through the following stages of compiler to translate the above program (iteratively) – all at once.**

Note: ***Only lines 5 (LET G = a + c), line 7 (M = A/B+C) and line 8 (N = G/H-I+a*B/c) should go through all the below seven (7) stages of compiler, since they have no errors.***
The other lines should just check for errors and report the error (if any)

OUR SEVEN STAGES OF COMPILER

1) Lexical Analysis (Scanner)
2) Syntax Analysis (Parser)
3) Semantic Analysis (Syntactic Analysis)
4) Intermediate Code Representation (ICR)
5) Code Generation (CG)
6) Code Optimization (CO)
7) Target Machine code (TMC) in Binary